INCIDENT RESPONSE

Initial Investigation

Confirming Real Incidents & Gathering Technical Context

AGENDA

01

Objective

Why initial investigation matters in incident response

02

Key Investigation Steps

Alert review, dashboards, host/app identification, false positive validation

03

Tools & Metrics

Splunk, ELK, cloud logs — CPU, memory, disk, network

04

Example Actions

Health checks, endpoint testing, synthetic transactions

05

Documentation

Incident ticket notes, screenshots, log excerpts

06

Summary & Best Practices

Key takeaways and recommendations

Why Initial Investigation?

Confirm whether the alert represents a real incident and gather the technical context necessary for an effective, targeted response.

🔍 Validate the Alert

Determine if the alert is genuine or a false positive caused by maintenance, expected spikes, or misconfiguration.

📊 Gather Context

Identify affected hosts, applications, networks, and databases to scope the impact accurately.

⏱️ Minimize Time-to-Resolve

Early, thorough investigation reduces Mean Time To Resolve (MTTR) and prevents escalation.

📝 Enable Collaboration

Documented findings allow faster handoffs and parallel workstreams across on-call teams.

4-Phase Investigation Framework

1 Alert Review

- Threshold breached
- Error codes
- Metric anomalies
- Alert priority / severity

2 Log & Dashboard Check

- Splunk queries
- ELK stack analysis
- Cloud provider logs
- APM traces

3 Impact Identification

- Affected hosts / VMs
- Application services
- Network segments
- Database instances

4 Change Correlation

- Recent deployments
- Patch history
- Config changes
- Maintenance windows

Monitoring Stack & Key Metrics to Check

Splunk

Log aggregation & correlation

ELK Stack

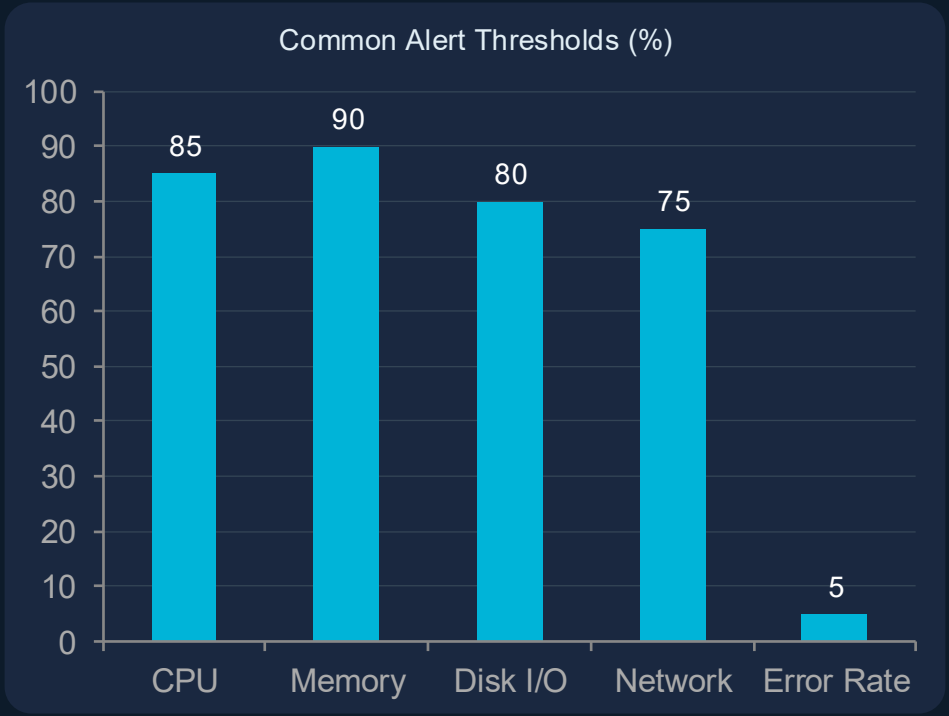
Elasticsearch, Logstash, Kibana

Cloud Logs

AWS CloudWatch / Azure Monitor / GCP Logging

APM Tools

Datadog, New Relic, Dynatrace



Is This a Real Incident?

Common False Positive Causes

Scheduled Maintenance Windows
Known downtime or restarts

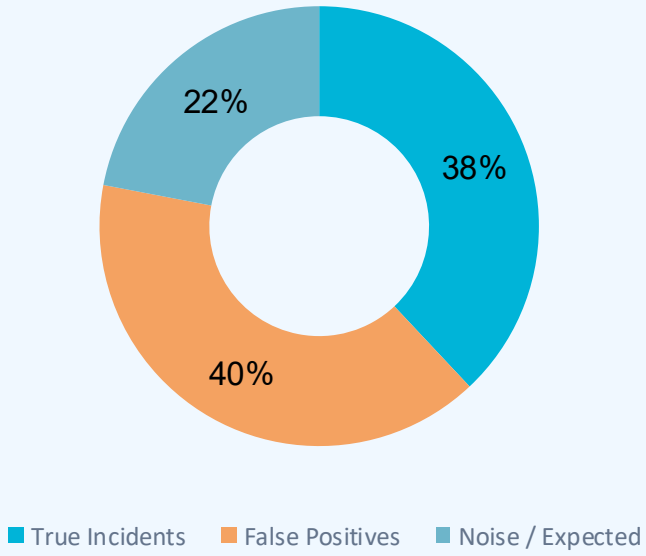
Expected Traffic Spikes
Seasonal / product launch surges

Configuration Changes
Threshold not updated after change

Auto-Scaling Events
Cloud scale-out triggering alerts

Load / Stress Testing
QA activities impacting prod metrics

Typical Alert Classification



EXAMPLE ACTIONS

Practical Steps During Investigation

Infrastructure Health

- → Run health checks on impacted servers (ping, SSH reachability)
- → Verify CPU utilization. Check for runaway processes
- → Monitor memory consumption and swap usage
- → Check disk I/O and available space on all mounts

Network & Connectivity

- → Verify network throughput and packet loss
- → Check firewall rules and security group changes
- → Trace network paths using traceroute / mtr
- → Review load balancer health and target group status

Application Layer

- → Test application endpoints (HTTP status codes)
- → Run synthetic transactions to simulate user flows
- → Review application error logs and stack traces
- → Check service dependencies and downstream APIs

Investigation Quality vs. Time to Resolve

~67% **MTTR Reduction**

For P1 incidents with structured initial investigation

40% **False Positives**

Average alert noise in enterprise environments

3× **Faster Escalation**

Pre-gathered context accelerates specialist handoffs

Capturing Investigation Evidence

INC-20240315-001 · Incident Ticket

Status:	In Progress — Initial Investigation
Severity:	P2 — High
Assigned:	On-Call Engineer
Started:	2024-03-15 09:42 UTC
Affected:	api-prod-01, api-prod-02
Alert:	HTTP 5xx Error Rate > 5% for 10 min

Investigation Notes:

09:44 — CPU normal. Memory at 92%. OOM killer triggered on api-prod-02.
09:48 — Deployment dc3a1f2 pushed at 09:30. Correlated with spike onset.

Documentation Checklist

- ✅ Timeline of events with timestamps
- ✅ Alert details: threshold, metric, value
- ✅ Affected component inventory
- ✅ Dashboard screenshots attached
- ✅ Relevant log excerpts copied
- ✅ Recent changes noted (deploy, patch)
- ✅ False positive assessment recorded
- ✅ Current status + next actions defined

Key Takeaways

01 **Don't skip false positive validation**

40%+ of alerts are noise, confirm before mobilizing resources.

02 **Time-box initial investigation**

Set a 15–30 minute window; escalate if unresolved.

03 **Check recent changes first**

Deployments & config changes are the #1 root cause category.

04 **Document in real time**

Don't reconstruct later, notes during investigation are most accurate.

05 **Use structured runbooks**

Pre-defined checklists prevent steps from being missed under pressure.

06 **Communicate early**

Even a brief status update prevents stakeholder escalation.

A thorough initial investigation is the foundation of every effective incident response. Invest the time upfront to save time overall.